EV Market Analysis

1. Project Overview

This project analyzes electric vehicle market data across leading manufacturers, vehicle models, pricing, driving range, customer ratings, and annual sales. The goal is to identify market trends, compare competitor performance, uncover underserved opportunities, and recommend Tesla's next strategic market segment using SQL, Python, and Power BI to support data-driven business decisions.

2. Dataset Summary

Dataset: twoPtPass.csv
- Rows: 2,001 - Columns: 24 - Key Features:
- Brand
- Model
- Year
- Variant
- Price
- Range
- Body Type
- Market Segment
- Annual Sales

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported dataset using `pandas`.
- **Initial Exploration:** Used `df.head()` to check structure.

	brand	model	year	variant	price_usd	battery_capacity_kwh	range_miles	charging_speed_kw	acceleration_0_60_mph	top_speed_mph	...	body_type
0	Volkswagen	ID. Buzz	2023	Performance	104880.80	118.7	400.0	234.5	3.04	195.0	...	Truck
1	Toyota	bZ Compact SUV	2022	Premium	48217.41	58.8	219.0	148.1	5.77	159.0	...	SUV
2	GM/Chevrolet	Bolt EV	2024	Premium	49651.12	58.2	225.0	104.9	6.84	148.0	...	Van
3	Kia	Sportage EV	2024	Long Range	38131.56	102.5	349.0	66.5	4.66	176.0	...	SUV
4	Tesla	Model 3	2022	Long Range	144079.87	93.9	314.0	298.5	5.64	165.0	...	SUV

5 rows × 24 columns

- **Data Cleaning:** Removed unnecessary columns, including null placeholder fields (battery_capacity_kwh, charging_speed_kw, drive_type, and top_speed_mph), to prepare them for analysis.
- **Featured Engineering:** Three additional columns were added to support the analysis: range_per_kwh to measure energy efficiency, price_per_mile to evaluate vehicle value, and estimated_revenue to estimate each model's market performance.

- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL

We performed structured analysis in PostgreSQL to answer key questions:

1. **Total Annual Sales by Brand** – Compared the total annual EV sales across all vehicle manufacturers.

	brand text	total_annual_sales numeric
1	Tesla	101459482
2	BYD	50472617
3	Volkswagen	48186657
4	Kia	16612129
5	Hyundai	15223045

2. **Average Price by Brand** – Compared the average vehicle price offered by each EV manufacturer.

	brand text	average_price numeric
1	Lucid	192516.88
2	Porsc...	157050.80
3	Merce...	128717.87
4	BMW	98062.34
5	Audi	93875.34

3. **Highest-Rated EV Models** – Identified the EV models with the highest average customer ratings.

	brand text	model text	average_customer_rating numeric
1	Merced...	EQC	3.86
2	Volvo	XC40 Rechar...	3.86
3	Merced...	EQS	3.83
4	Porsche	Taycan	3.83
5	Merced...	GLE EV	3.82

4. **Average Driving Range by Brand** – Compared the driving range offered by each EV manufacturer.

	brand text	average_range numeric
1	Honda	326.00
2	Porsche	274.86
3	Mercedes	270.79
4	Toyota	269.66
5	Rivian	268.70

5. **Sales by Market Segment** – Compared total annual sales across different EV market segments.

	market_segment text	total_sales numeric
1	Premium	86507799
2	Mid-range	83479890
3	Luxury	65966088
4	Budget	23353407

6. **Best Range per Dollar**– Identified the EV models that provide the greatest driving range for their price.

	brand text	model text	price_usd double precision	range_miles double precision	range_per_dollar numeric
1	BYD	Qin	17777.32	280	0.0158
2	BYD	Seagull	19470.99	291	0.0149
3	BYD	Seagull	16613.01	237	0.0143
4	BYD	Qin	27057.77	385	0.0142
5	BYD	Qin	21712.4	307	0.0141

7. **Average Charging Speed by Brand**– Compared the average charging speeds offered by each EV manufacturer.

	brand text	average_charging_speed numeric
1	Porsche	279.27
2	Tesla	276.97
3	Lucid	259.28
4	Honda	166.50
5	Volvo	139.08

8. **Customer Ratings by Market Segment** - Compared the average customer ratings across EV market segments.

	market_segment text	average_customer_rating numeric
1	Luxury	3.70
2	Premium	3.55
3	Mid-range	3.53
4	Budget	3.45

9. **Sales by Price Range -** Compared total annual sales across different EV price ranges.

	price_range text	total_sales numeric
1	$70k-$99k	72620150
2	$100k+	65966088
3	$50k-$69k	55485755
4	$30k-$49k	53200462
5	Under $30k	12034729

10. **Average Driving Range by Body Type -** Compared the average driving range across different vehicle body types.

	body_type text	average_range numeric
1	Coupe	267.95
2	Truck	264.39
3	SUV	260.76
4	Van	259.43
5	Sedan	259.08

11. **Brand Performance by Market Segment -** Compared annual sales by brand within each EV market segment to identify market leaders.

	market_segment text	brand text	total_sales numeric
1	Budget	BYD	12646894
2	Budget	Tesla	4563079
3	Budget	Volkswagen	2705160
4	Budget	Kia	1813751
5	Budget	Hyundai	1112878

12. **Market Segment Opportunity Analysis -** Evaluated market segments based on sales, customer ratings, and competition to identify Tesla's strongest growth opportunity.

	market_segment text	total_sales numeric	avg_rating numeric	number_of_models bigint
1	Premium	86507799	3.55	69
2	Mid-range	83479890	3.53	63
3	Luxury	65966088	3.70	62
4	Budget	23353407	3.45	36

13. **Body Type Opportunity Analysis -** Evaluated vehicle body types based on sales, customer ratings, and competition to identify the best opportunity for Tesla.

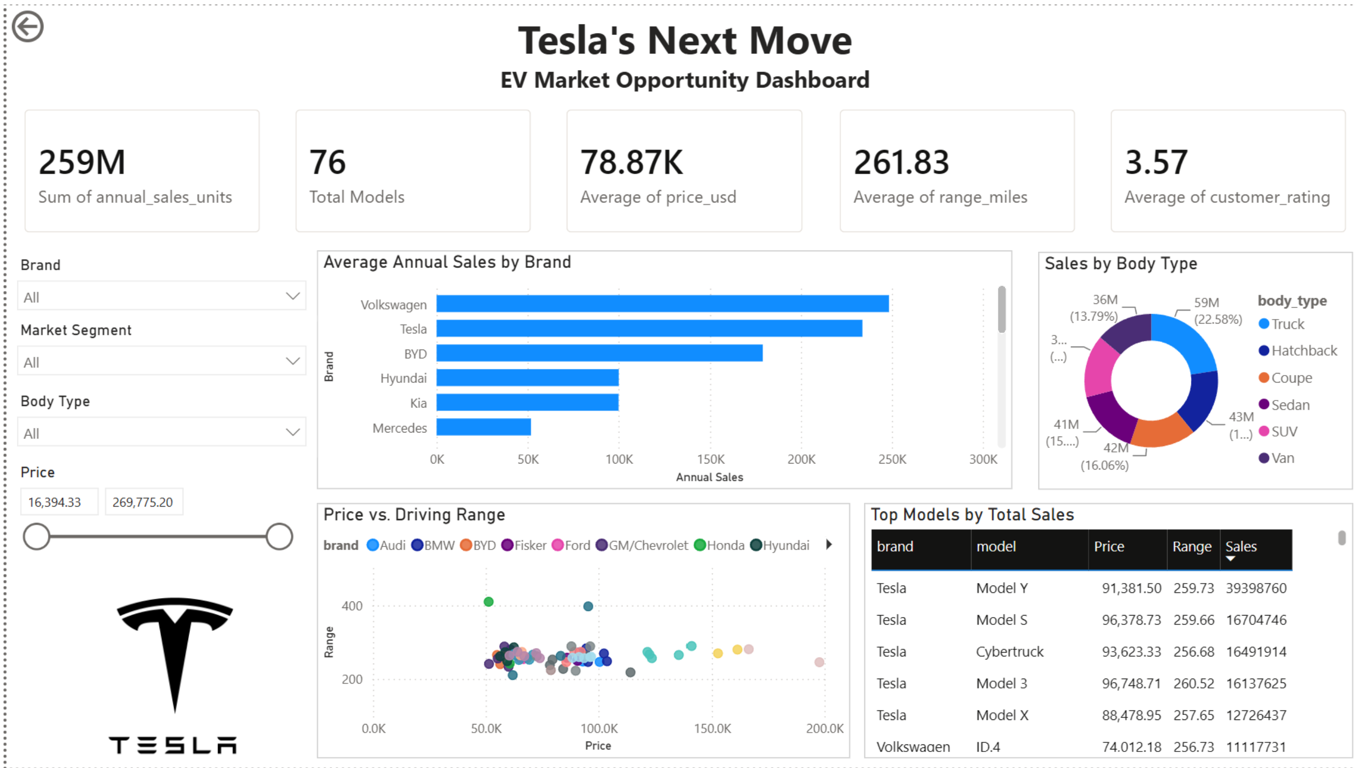
	body_type text	total_sales numeric	average_rating numeric	number_of_models bigint
1	Truck	58540982	3.61	63
2	Hatchback	42792674	3.56	60
3	Coupe	41644598	3.60	58
4	Sedan	41159954	3.55	58
5	SUV	39409797	3.54	66

14. **Price Range Opportunity Analysis** - Evaluated EV price ranges based on sales, customer ratings, and market competition to determine Tesla's optimal target price segment.

	price_range text	total_sales numeric	average_rating numeric	number_of_models bigint
1	$70k-$99k	72620150	3.55	69
2	$100k+	65966088	3.70	62
3	$50k-$69k	55485755	3.53	64
4	$30k-$49k	53200462	3.51	45
5	Under $30k	12034729	3.46	22

5. Dashboard in Power BI

Finally, we built a dashboard in **Power BI** to present insights visually.



6. Recommendations

- **Expand into the Premium Market Segment** – The Premium segment generated the highest annual sales while maintaining strong customer ratings, making it the most attractive market for future Tesla expansion.

- **Focus on the $70k–$99k Price Range -** This price range recorded the highest total sales, indicating strong consumer demand for premium EVs. Tesla should continue developing vehicles that offer competitive performance within this segment.

- **Target High-Demand Body Types** – Body types such as trucks and hatchbacks demonstrated strong sales and customer ratings, suggesting potential opportunities for expanding Tesla's product lineup beyond its current offerings.